

# Oikos

## System design summary: home health intake agent

|            |                                                                                              |
|------------|----------------------------------------------------------------------------------------------|
| Event      | Healthcare AI Hackathon (AWS, San Francisco), 26 September 2026                              |
| Team       | Sankalpa Venkatraghavan, Terry Lin, Himani Shah                                              |
| Repository | <a href="https://github.com/sankalpav/Oikos">github.com/sankalpav/Oikos</a>                  |
| Stack      | Python, Streamlit, GPT-5.6 Terra on Amazon Bedrock (OpenAI-compatible Responses API), pandas |
| Data       | 100% synthetic. No real patient information was used at any point.                           |
| Status     | Working demo; live model verified end to end; cached fallback for offline use                |

**One line:** every home health referral gets a next step. Accepted patients are billed fully; declined referrals go back to the hospital with a note of next steps.

**Why the name:** *oikos* is Greek for household or home, the place where both home health and home care happen.

## 1. Executive summary

Hospitals and skilled nursing facilities (SNFs) send discharge referrals to home health agencies, mostly through CarePort, fax and phone. Agencies screen these by hand, respond slowly, and now accept only around a third of referrals, mostly because they cannot staff them. Of the patients they do accept, the coder sees only a short intake-nurse summary, so documented comorbidities never reach the claim. Patients they decline receive no next step, even when they qualify for non-medical home care programs.

Oikos is an intake agent that an agency subscribes to. For each referral it (1) extracts clinical facts from the discharge packet with an LLM, (2) applies deterministic rules for insurance, eligibility, paperwork and nurse capacity to decide accept / fixable / decline, then (3a) for accepted patients surfaces missed billable diagnoses with chart evidence, or (3b) for declined referrals checks home care programs and adds a rejection note to the referral file for the hospital's discharge planner. Oikos never contacts patients or families.

The key strategic decision was that **nurses, not patients, are the bottleneck**. Oikos is therefore positioned as maximizing revenue per nurse and giving every referral an outcome, rather than simply generating more referrals.

## 2. Problem context

### 2.1 How a referral flows today

- A hospital or SNF case manager creates a discharge referral and broadcasts it to many agencies via CarePort, fax or phone.
- An agency intake coordinator reads the packet by hand and checks insurance, homebound status, skilled need and the face-to-face encounter documentation.
- The coordinator checks whether a qualified clinician can reach the patient's zip code within the 48-hour start-of-care window required by Medicare.
- The first agency to accept usually wins the patient (team research: roughly 21% higher assignment rate for the first responder).

- An RN performs the start-of-care visit and OASIS assessment, writes a short summary, and sends it to a coder, who bills a 30-day PDGM payment period using only that summary.
- Declined referrals are dropped; the patient goes elsewhere or receives nothing.

## 2.2 Market and bottleneck statistics used

| Statistic                                     | Value                            | Source                                                   | Confidence   |
|-----------------------------------------------|----------------------------------|----------------------------------------------------------|--------------|
| Medicare-certified home health agencies       | 12,000+                          | MedPAC, Dec 2025 (2024 data)                             | High         |
| Traditional Medicare home health users / yr   | 2.7M                             | MedPAC, Dec 2025                                         | High         |
| 30-day payment periods / yr                   | 8.3M                             | MedPAC, Dec 2025                                         | High         |
| Traditional Medicare home health spend / yr   | \$16.0B                          | MedPAC, Dec 2025                                         | High         |
| Medicare margin vs all-payer margin           | 21.2% vs 5.0%                    | MedPAC, Dec 2025                                         | High         |
| Referral acceptance rate                      | ~30% (from ~75-80% pre-pandemic) | Luna blog (vendor), citing industry data                 | Medium       |
| Providers turning away referrals for staffing | 77%                              | 2023 provider survey, cited by Luna and others           | Medium       |
| Home health referrals, year over year         | +4.6%                            | WellSky network data via Home Health Care News, Dec 2025 | High         |
| Comorbidity burden (Van Walraven) since 2019  | +34%                             | WellSky via Home Health Care News, Dec 2025              | High         |
| First-responder assignment advantage          | ~21%                             | Team discovery research (not independently sourced)      | Low - verify |

## 2.3 Home health vs home care

|                | Home health (skilled)                                     | Home care (non-medical)                                 |
|----------------|-----------------------------------------------------------|---------------------------------------------------------|
| Services       | Nursing, PT/OT/SLP, wound care, IV, medication management | Bathing, dressing, meals, housekeeping, companionship   |
| Order required | Yes: physician certifies, plan of care                    | No                                                      |
| Eligibility    | Homebound + intermittent skilled need                     | Anyone who needs help                                   |
| Main payers    | Medicare (FFS and Advantage), Medicaid, commercial        | Private pay, LTC insurance, Medicaid waivers / IHSS, VA |
| Role in Oikos  | The paying customer's core business                       | Fallback route for declined patients                    |

## 3. Key decisions and rationale

| Decision            | Options considered                   | Choice                                     | Rationale                                                                                              |
|---------------------|--------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Bottleneck to solve | More patients vs more nurse capacity | Nurse capacity; maximize revenue per nurse | Agencies already decline most referrals; 77% cite staffing. More referrals alone does not add revenue. |

| Decision           | Options considered                                                           | Choice                                                               | Rationale                                                                                                                           |
|--------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Customer           | Home health agencies (B2B), families (D2C), hospitals                        | Sell to agencies (B2B)                                               | Agencies have budget and a clear ROI. D2C has high acquisition cost, and agencies may not contact declined patients directly.       |
| Patient contact    | Text families a care plan vs note on the referral file                       | Rejection note on the referral file only                             | Agencies may not contact declined patients; only the referring hospital communicates with the patient.                              |
| Integration path   | Integrate with CarePort directly, or act on the agency's behalf              | Agent runs on each agency's own portal access                        | CarePort requires an NPI and has no preference logic (first come, first served); an agency-side agent avoids needing a partnership. |
| Revenue directions | 1) reroute rejected patients, 2) source from SNFs/LTC, 3) billing and coding | Demo builds 1 and 3 on top of triage; 2 on the roadmap               | SNF sourcing relies on phone/fax relationships and is not demoable in an hour; the other two are self-contained.                    |
| Hackathon scope    | Triage only, triage + one layer, all three                                   | All three layers (team choice)                                       | Shows the full story: every referral gets an outcome.                                                                               |
| LLM vs rules split | End-to-end LLM decisioning vs LLM extraction + deterministic rules           | LLM for reading and writing; rules for every decision                | Decisions must be auditable and repeatable; eligibility and staffing are rule-shaped; LLMs are strong at unstructured text.         |
| Model              | Claude via Bedrock, GPT-5.6 Terra via Bedrock                                | GPT-5.6 Terra (team choice)                                          | Matches the AWS workshop setup already configured; OpenAI-compatible Responses API.                                                 |
| Resilience         | Live only vs live with fallback                                              | Live with per-step cached fallback                                   | Workshop credentials expire in ~24 hours and demos cannot depend on Wi-Fi.                                                          |
| UI                 | Streamlit, Next.js, CLI                                                      | Streamlit                                                            | Fastest path to a demoable multi-view app in Python; free public hosting.                                                           |
| Billing safety     | Auto-code vs surface evidence                                                | Surface evidence only; clinician confirms, coder codes               | Avoids upcoding risk; keeps clinical judgment with the nurse.                                                                       |
| First market       | National vs one state                                                        | California                                                           | IHSS pays family members as caregivers, a strong option to surface for discharge planners; team is local.                           |
| Pricing            | Per-referral or percentage-of-revenue vs flat fee                            | Flat per-branch software fee (+ private-pay home care referral fees) | Referral-linked payments raise federal anti-kickback concerns, especially for Medicaid/IHSS-funded patients.                        |
| Hosting            | GitHub Pages, Streamlit Community Cloud, laptop tunnel                       | Streamlit Community Cloud from the GitHub repo                       | GitHub Pages is static only; Community Cloud is free and runs Python. Public link runs cached mode so no one can spend AWS credits. |
| Name               | Constant Care (placeholder), Baton, Census, Homebound, others                | Oikos                                                                | Team choice; means household or home.                                                                                               |

## 4. System architecture

## 4.1 Pipeline

```
Referral (structured header + discharge packet text)
|
v
[1] LLM extraction -----> diagnoses + ICD-10 + exact quotes, homebound, skilled need,
|                             services (discipline + skills), face-to-face, ADL/IADL needs
v
[2] Rules: triage -----> insurance | skilled need | homebound | face-to-face | staffing
|                             => ACCEPT / FIXABLE / DECLINE (+ reason)
+-- ACCEPT / FIXABLE --> [3a] LLM billing capture (missed dx vs nurse summary)
|                       + templated fix actions (face-to-face request, prior auth)
+-- DECLINE -----> [3b] Rules: home care qualification + agency match
|                       + LLM rejection note on the referral file (for discharge planner)
v
Quote verification: every evidence quote is checked verbatim against the source text
```

## 4.2 Components

| File        | Responsibility                                                                                                                                                                                                            |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| app.py      | Streamlit UI: Overview (problem statistics), Referral queue, Referral detail, Impact (metrics and ROI sliders). Sidebar toggles live model, tests the Bedrock connection, and runs all referrals in parallel (5 threads). |
| pipeline.py | Prompts and schemas, triage rules, nurse matching, fix actions, home care rules, orchestration, quote verification, and mock fallback per step.                                                                           |
| llm.py      | Bedrock client using the OpenAI SDK against the Bedrock OpenAI-compatible endpoint. Uses OPENAI_API_KEY if set, otherwise generates a short-lived Bedrock token from standard AWS credentials. Robust JSON parsing.       |
| data/       | Synthetic referrals, nurse roster, home care agencies, payer table, zip centroids, PACE areas, cached model outputs.                                                                                                      |

## 4.3 LLM steps

| Step            | Input                                                                                     | Output (JSON)                                                                                                                            | Guardrails in prompt                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extraction      | Discharge packet; allowed skill vocabulary; allowed ADL/IADL list                         | summary, primary_diagnosis, diagnoses[], homebound, skilled_need, services[], face_to_face_documented, adl_needs[], cognitive_impairment | Use only stated facts; every evidence field is an exact quote; home health aftercare/sequela ICD-10 conventions; at most 2 essential skills per service          |
| Billing capture | Discharge summary, nurse summary, extracted diagnoses                                     | missed[] (name, icd10, evidence, why_it_matters), coding_notes[], nurse_query                                                            | Exclude conditions already mentioned in the nurse summary, even by abbreviation; apply ICD-10 'with' conventions; never invent; clinician confirms               |
| Rejection note  | Failed checks, staffing detail, program results, matched agencies, skilled follow-up flag | summary, note_markdown                                                                                                                   | Written for the discharge planner, never the patient; reason, what would change the decision, skilled hand-off, home care options; 'likely' / 'may qualify' only |

## 4.4 Triage rules

| Check        | Pass                                                | Fixable                                                           | Fail (decline)                        |
|--------------|-----------------------------------------------------|-------------------------------------------------------------------|---------------------------------------|
| Insurance    | Contracted payer, no prior auth                     | Contracted payer requiring prior auth (drafts request)            | Payer not contracted / out of network |
| Skilled need | Documented skilled nursing or therapy need          | -                                                                 | No skilled need                       |
| Homebound    | Documented taxing effort / assistance to leave home | -                                                                 | Not homebound                         |
| Face-to-face | Encounter documented                                | Missing (drafts request to discharge planner, holds slot 4 hours) | -                                     |
| Staffing     | Every required discipline covered in time           | -                                                                 | Any discipline uncovered              |

Decision logic: any **fail** gives DECLINE (reason = first failing check); otherwise any **fixable** gives FIXABLE; otherwise ACCEPT.

#### 4.5 Nurse matching algorithm

- For each required service, candidates are clinicians whose license equals the discipline (RN, PT, OT, SLP). LVNs cannot open a case.
- The start-of-care discipline is RN if ordered, otherwise the first therapy; it must be seen within 48 hours of discharge. Other disciplines must be seen within 5 days.
- A clinician qualifies if they have all required skills and the patient is within their travel radius (haversine distance between zip centroids).
- Among qualified clinicians, the earliest open slot before the deadline is chosen. If none meets the deadline, the nearest qualified clinician and their first slot are shown as the reason for declining.

#### 4.6 Home care qualification rules (California-focused, general program rules)

| Program                  | Rule used in demo                                                                                  | Status shown                                    |
|--------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------|
| IHSS                     | On Medi-Cal and at least one ADL/IADL need                                                         | Likely; family member can be the paid caregiver |
| Medi-Cal MSSP            | On Medi-Cal, age 65+, and 2+ ADLs or cognitive impairment                                          | Possible                                        |
| VA Aid & Attendance      | Wartime veteran or surviving spouse, and at least one ADL need                                     | Possible (income and asset limits apply)        |
| PACE                     | Age 55+, zip in a PACE service area, and 2+ ADLs or cognitive impairment                           | Possible                                        |
| Long-term care insurance | Has a policy, and 2+ ADLs or cognitive impairment (common benefit trigger)                         | Likely                                          |
| Private pay              | Agencies serving the zip; hours/week = 4 per ADL + 2 per IADL (cap 40); cost = rate x hours x 4.33 | Always listed                                   |

If the patient still needs skilled care and was declined for staffing or insurance, Oikos adds a warning to notify the discharge planner immediately so another agency can accept the skilled referral. Home care never replaces skilled care.

#### 4.7 Impact and ROI model

- Average payment per 30-day period = \$16.0B / 8.3M periods = about **\$1,928** (MedPAC 2024 data).
- Periods per admission = **1.5** (assumption).

- Recovered admissions / month = referrals x (1 - acceptance rate) x share of declines that are insurance-eligible x recovery rate.
- Added revenue / year = recovered admissions x \$1,928 x 1.5 x 12. Billing-capture uplift is excluded.
- Home care referral fees = declines x share placed in home care x fee x 12 (private pay only).
- Defaults: 200 referrals/month, 35% acceptance, 45% eligible, 15% recovered, 20% placed, \$300 fee; about \$305K/year added revenue. All adjustable.

## 5. Synthetic data: how it was created and sourced

All patient-level data was written by hand for this demo (drafted with Claude and reviewed by the team). Nothing was copied from real records. Every facility, payer (other than Original Medicare), clinician, agency and PACE program name is fictional and labeled (*synthetic*) in the app. The goal was a small set of referrals where each one exercises a different branch of the system, with realistic clinical language and deliberately planted gaps.

### 5.1 Design principles

- **One scenario per branch:** accept, fixable, and three different decline reasons (staffing, clinical, insurance).
- **Planted billing gaps:** discharge summaries document comorbidities that the short nurse summary omits, mirroring the real problem that coders see only the nurse's summary.
- **Realistic structure:** discharge summaries follow common sections (principal diagnosis, hospital course, past history, functional status, disposition, home health orders, face-to-face encounter, follow-up).
- **Homebound and eligibility language** is written the way Medicare documentation typically phrases it (e.g. 'leaving home requires a considerable and taxing effort').
- **Consistent time frame:** all discharges on 26 Sep 2026, so the 48-hour start-of-care deadline is meaningful against the roster's open slots.
- **Real geography, fake people:** San Francisco Bay Area zip codes with approximate real centroid coordinates, so distances are plausible.

### 5.2 The five referrals

| ID      | Patient (synthetic)                                                        | Scenario                                                    | Planted features                                                                                                               | Expected outcome                                                            |
|---------|----------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| HH-1001 | Eleanor Park, 78F, 94110, Original Medicare                                | Hip fracture after fall, s/p ORIF (hospital)                | Nurse summary omits type 2 diabetes with neuropathy on insulin, CKD 3b, recurrent depression; HTN + CKD coding convention      | ACCEPT; RN, PT and OT matched; 5 missed diagnoses                           |
| HH-1002 | Walter Brooks, 82M, 94601, Medicare Advantage (synthetic)                  | HFpEF exacerbation (hospital)                               | Prior auth required; face-to-face missing; nurse summary omits COPD, AFib on apixaban, stage 2 sacral pressure injury, obesity | FIXABLE; face-to-face and prior auth drafts; 6-7 missed diagnoses           |
| HH-1003 | Rosa Delgado, 86F, 95020 Gilroy, Original Medicare + Medi-Cal              | Stage 3 heel ulcer on wound vac, moderate Alzheimer's (SNF) | Only wound-vac-trained RN within radius is first free 73 hours after discharge                                                 | DECLINE (staffing); IHSS + MSSP in rejection note; skilled hand-off warning |
| HH-1004 | Harold Jensen, 79M, 94501, Original Medicare; Vietnam veteran; lives alone | Pneumonia, resolved (hospital)                              | Independent 300 ft, explicitly not homebound; needs help bathing, meals, housekeeping                                          | DECLINE (clinical); VA Aid & Attendance; 2 private-pay agencies             |

| ID      | Patient (synthetic)                                                          | Scenario                                                 | Planted features                                                       | Expected outcome                                                    |
|---------|------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------|
| HH-1005 | Linda Chen, 62F, 94112, employer PPO (synthetic, out of network); LTC policy | Right MCA stroke, left hemiparesis, dysphagia (hospital) | Payer not contracted, although the agency could staff every discipline | DECLINE (insurance); LTC insurance + PACE; skilled hand-off warning |

### 5.3 Supporting data files

| File                            | Contents                                                                              | How it was made                                                                                                                    |
|---------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| nurses.csv                      | 9 clinicians: license, skills, home zip, travel radius, next open slot, visits left   | Hand-authored so each referral has a clear staffing result (e.g. a wound-vac RN 9.8 miles away but outside the 48-hour window).    |
| homecare_agencies.csv           | 4 private-pay agencies: zips served, hourly rate (\$34-40), LTC acceptance, languages | Fictional; rates chosen to be in a plausible Bay Area range for illustration only.                                                 |
| reference.json: payers          | Original Medicare, a synthetic Medicare Advantage plan, a synthetic employer PPO      | Contract and prior-auth flags set to create the fixable and out-of-network scenarios.                                              |
| reference.json: zips            | 10 Bay Area zip codes with approximate centroid lat/lon and county                    | Approximate public zip centroids; used only for distance and county lookup.                                                        |
| reference.json: PACE areas      | Two fictional PACE programs mapped to zip lists                                       | Illustrative; real PACE service areas differ and must be looked up.                                                                |
| referrals.json: manual baseline | Per referral: manual decision, minutes to decision, note                              | Assumed values (150-240 minutes) representing today's manual process; not measured data.                                           |
| data/mock/*.json                | Cached extraction and billing outputs per referral                                    | Written by hand to match the expected model output; every quote copied verbatim from the synthetic summaries and verified by code. |

### 5.4 Clinical coding notes

ICD-10 codes in the synthetic data and cached outputs follow standard home health conventions (e.g. fracture aftercare with 7th character D, stroke sequela I69.x, diabetes-CKD and hypertension-CKD/heart failure 'with' presumptions). They are illustrative and must be reviewed by a certified coder before any real use.

## 6. Testing and validation

- **Rules in cached mode:** all five referrals produce the expected decision and reason; every cached quote verifies against its source text.
- **UI:** Streamlit AppTest renders every tab and every referral with no exceptions.
- **Live model (GPT-5.6 Terra on Bedrock):** all five decisions matched the design. Per-referral latency about 8-24 seconds; parallel run of all referrals about 20 seconds (down from about 75 seconds sequentially).
- **Rejection notes:** live notes for all three declines state the reason, what would change it, the skilled hand-off and home care options.
- **Guardrail observed:** in one live run the model paraphrased an evidence quote; quote verification flagged it in the UI.

### 6.1 Fixes made after live testing

| Issue found                                                                                                                   | Fix                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Billing step listed the primary diagnosis as 'missed' although the nurse summary mentioned it by abbreviation (CHF, hip ORIF) | Prompt now excludes conditions already mentioned, even by abbreviation, unless the code should change |
| Early versions texted families a care plan                                                                                    | Replaced with a rejection note on the referral file; agencies may not contact declined patients       |
| Run-all took about 75 seconds                                                                                                 | Parallelized with a 5-thread pool                                                                     |
| Dollar signs rendered as LaTeX math in Streamlit captions                                                                     | Dollar signs escaped in captions and model-generated notes                                            |
| An unsourced '20-40 page packet' claim on the Overview tab                                                                    | Removed; replaced with 'long discharge packets'                                                       |

## 7. Guardrails, compliance and security

- **No real PHI:** synthetic data only. A production version needs a BAA with AWS, Bedrock data-handling review and HIPAA controls.
- **Clinical safety:** Oikos accepts only when a real clinician slot exists inside the 48-hour start-of-care window; declined skilled patients trigger a hand-off warning.
- **Coding integrity:** suggestions show chart quotes and require clinician confirmation; nothing is auto-coded or auto-billed.
- **Anti-kickback:** no payments tied to hospital referrals; flat software pricing; home care referral fees only for private pay, pending healthcare counsel review. IHSS is consumer-directed, so there is usually no agency to charge.
- **Eligibility language:** 'likely' / 'possible' only; the county, VA or insurer decides.
- **Credentials:** AWS keys are never stored in the repo; locally they live in a temporary session file; on Streamlit Cloud they would go in Secrets. The public link runs cached mode by default so no one can spend AWS credits.

## 8. Risks and open questions

| Risk / question                                          | Why it matters                                                       | Next step                                            |
|----------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------|
| CarePort automation and terms of service                 | No API; the agent depends on agency credentials and could be blocked | Review terms; explore official partner or EHR routes |
| 21% first-responder statistic                            | Central to the speed story but not independently sourced             | Find a primary source or measure with a pilot agency |
| Acceptance and staffing figures come from vendor sources | Judges and buyers may challenge them                                 | Replace with WellSky, NAHC or peer-reviewed sources  |
| State-by-state program rules                             | Eligibility rules differ outside California                          | Build a per-state rules table                        |
| PDGM payment tables not implemented                      | No dollar value per missed diagnosis                                 | Add PDGM comorbidity and clinical group logic        |
| Real roster and scheduling data                          | Matching is only as good as the schedule data                        | Integrate with agency EMRs (e.g. WellSky, HCHB)      |

## 9. Roadmap

| Phase           | Scope                                                                                  |
|-----------------|----------------------------------------------------------------------------------------|
| Now (hackathon) | Triage + billing capture + home care qualification on synthetic data; public demo link |

| Phase | Scope                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------|
| Next  | Pilot with 1-3 California agencies; real portal ingestion (CarePort, fax OCR, Epic); EMR roster sync; PDGM logic                |
| Later | SNF/LTC referral sourcing; discharge-planner tools; Medicare Advantage / ACO contracts; more states; standalone coding platform |

## Appendix: running the demo

```
cd "09 Hackathons/aws/oikos"
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py
```

```
# live model: export AWS workshop credentials in the same terminal first
export AWS_REGION=us-east-1 AWS_ACCESS_KEY_ID=... AWS_SECRET_ACCESS_KEY=... AWS_SESSION_TOKEN=...
```

Sources: MedPAC home health update (Dec 2025); Home Health Care News on WellSky referral data (Dec 2025); Luna, 'Why home health referral acceptance rates are collapsing'; McKnight's Home Care; Hospice News and STAT on the UnitedHealth-Amedisys transaction. Program rules are general summaries and must be verified with official sources.